ProjectNameHere

2D Metroidvania — Unity 6 — Full Project Architecture

Generated 2026-02-21 · 155 Scripts · 62 Beads · 9 Agents

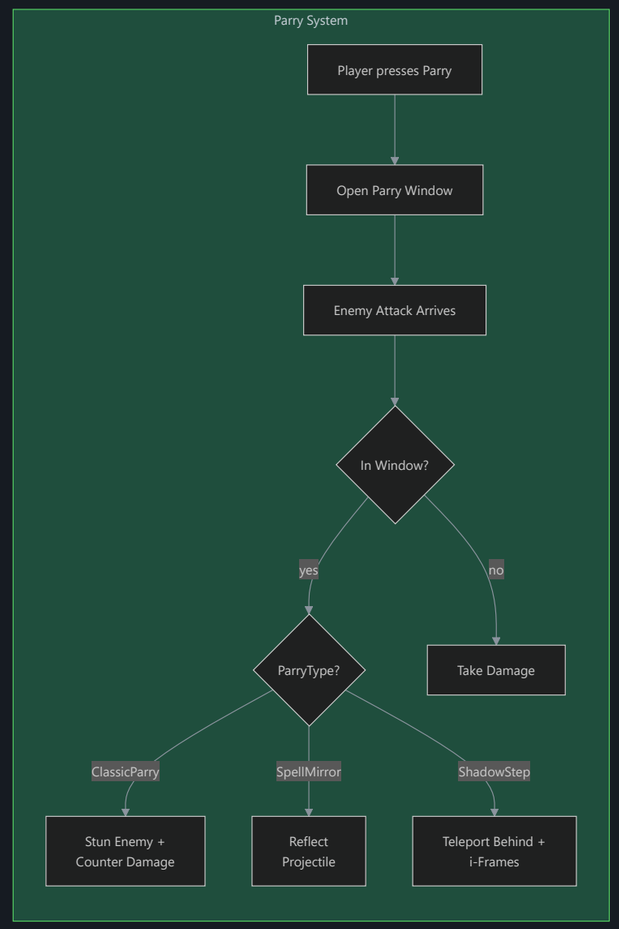
155	16	9	62	49	7	6	0
SCRIPTS	SYSTEMS	AGENTS	BEADS	CLOSED	IN PROGRESS	BLOCKED	READY

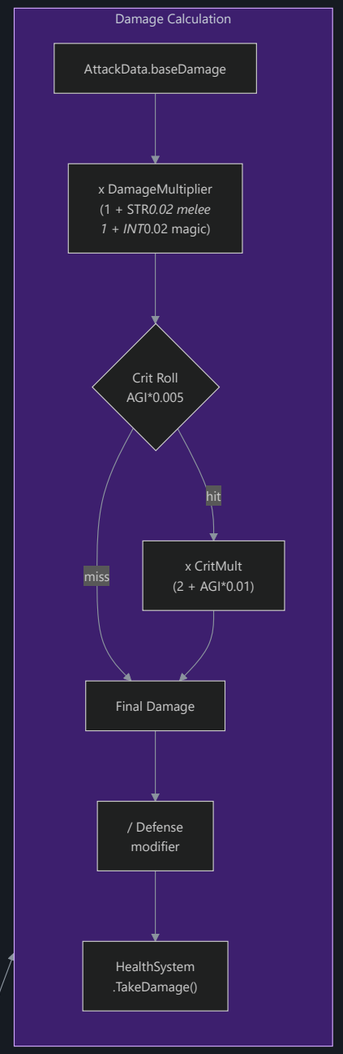
1. High-Level System Map

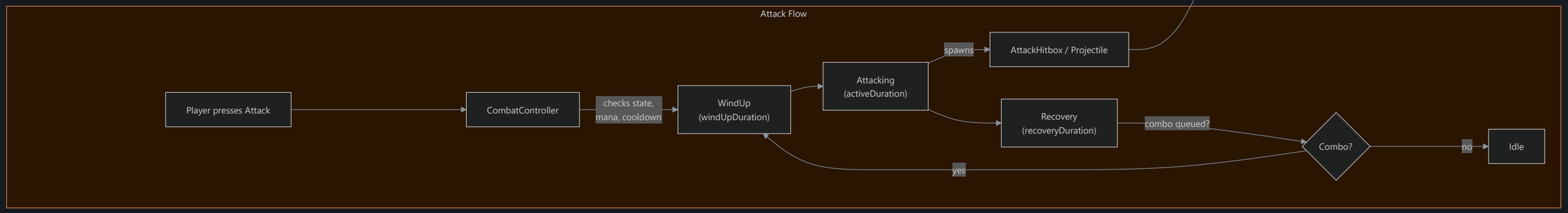
Every major system in the game and how they connect.

2. Combat System Detail

Attack flow, parry mechanics, combo chains, and damage calculation.







Weapon Slots

Key Constants

- **Melee** — Directional (Forward/Up/Down) + aerial overrides
- **Ranged** — Spawns Projectile with pierce
- **Magic** — Mana-consuming, can spawn SkillProjectile

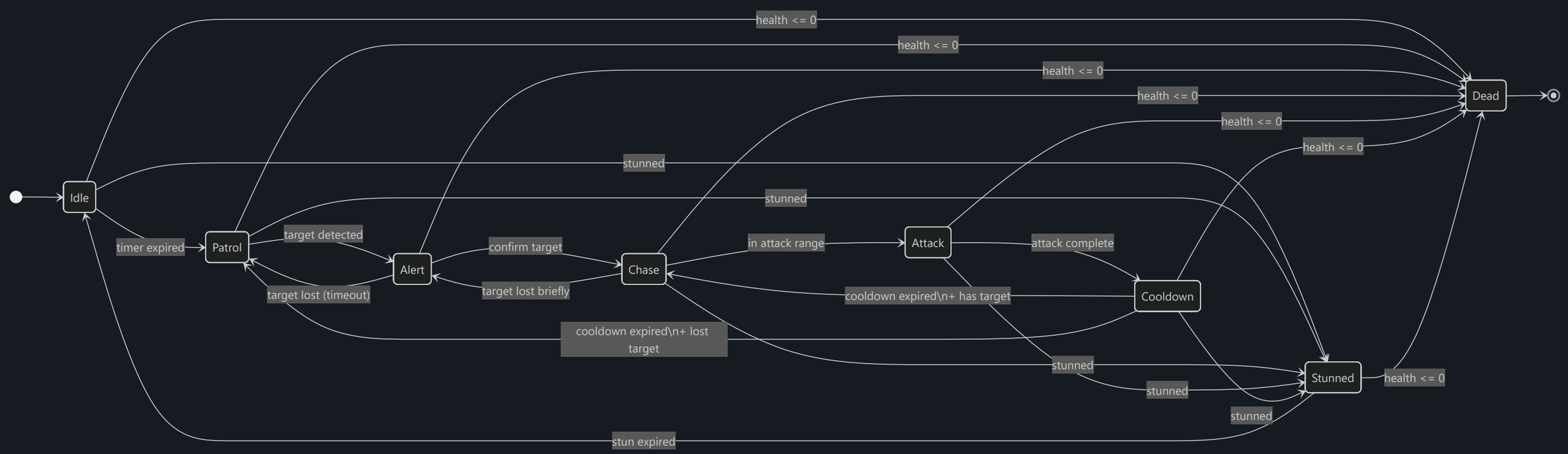
Combat States

- Idle → WindUp → Attacking → Recovery → (Combo?) → Idle
- Parrying (entered from Idle on parry input)

Param	Value
Coyote Time	0.15s
Jump Buffer	0.15s
Fall Gravity Mult	2.5x
Max Fall Speed	-20
Dash Speed / Duration / CD	20 / 0.2s / 1.0s
Crit Cap	50% (AGI*0.005)

3. Enemy AI & Wave System

State machines, boss phases, sensor detection, and wave progression.



Boss Phase System

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mermaid version 11.12.3

Wave Lifecycle

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mermaid version 11.12.3

Enemy Types

- **GroundPatrol** — Walk, wall/ledge flip

Detection Modes

- **Radius** — Sphere check around enemy

Enemy Prefabs

- Bat, Flying Eye

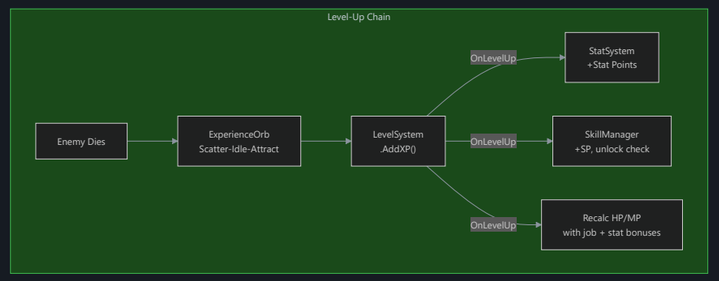
- **Flying** — No gravity, 2D flight
- **Hopping** — Jump cycles, hop cooldowns
- **Stationary** — Turrets, no movement

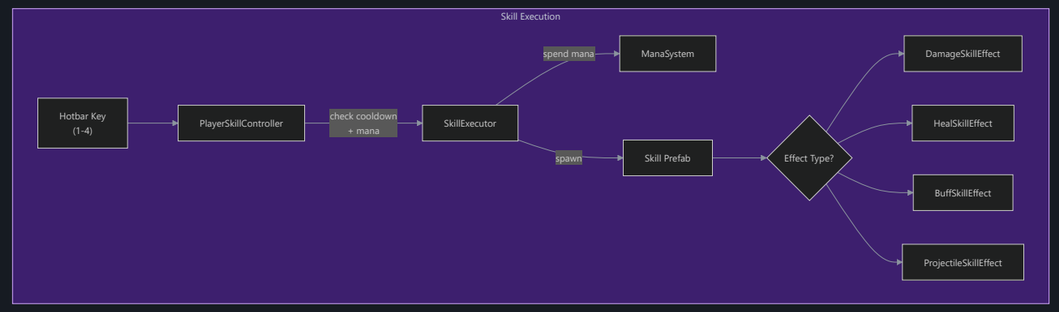
- **Cone** — Angle + range in facing dir
- **LineOfSight** — Raycast, blocked by walls

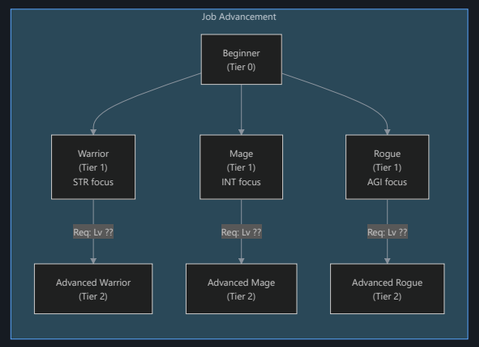
- Goblin, Skeleton
- Mushroom, Slime
- Turret
- GuardianBoss

4. Skills, Jobs & Progression

Job advancement tree, skill system, and leveling formulas.







Stat Formulas

Stat	Formula
Max HP	`baseHP + (lvl-1)*hpPerLvl + job.hpBonus + STR*5`
Max Mana	`baseMana + (lvl-1)*mpPerLvl + job.mpBonus + INT*3`
Melee Mult	`1 + STR * 0.02`
Magic Mult	`1 + INT * 0.02`
Atk Speed	`1 / (1 + AGI * 0.01)`
Crit Chance	`min(0.5, AGI * 0.005)`
Crit Damage	`2 + AGI * 0.01`
XP to Level	`floor((L + 300 * 2^(L/7)) / 4)`

Skill Data Model

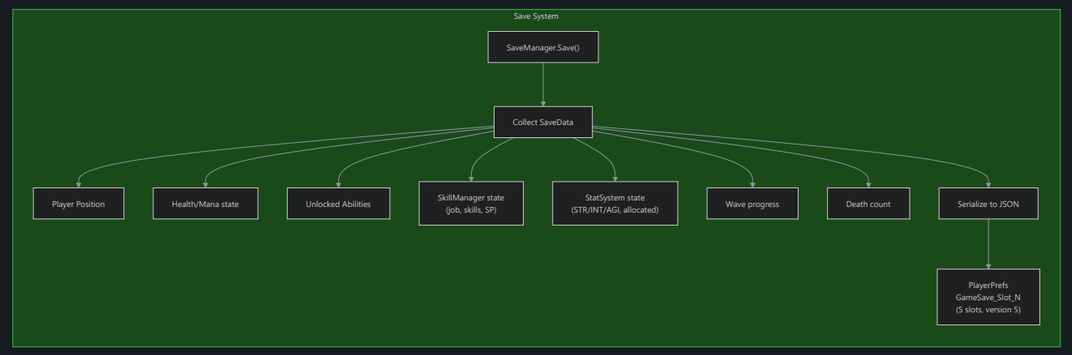
- **SkillData (SO)** — id, name, type, damage type, requirements, scaling per level, effects[], prefab, sound, animation
- **JobClassData (SO)** — job tree, stat modifiers, growth rates, parryData, appearance, available skills
- **SkillTreeData (SO)** — hierarchy of skill nodes, connections
- **SkillEffectData (SO)** — individual effect template (damage/heal/buff/projectile)
- **SkillInstance** — runtime state: current level, cooldown tracker

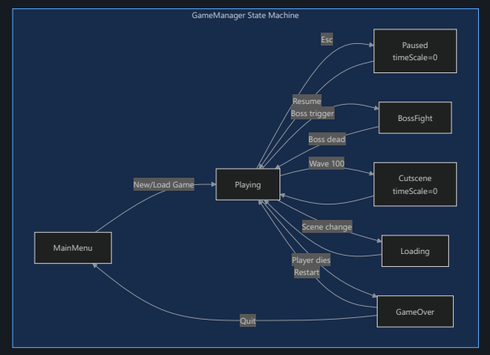
Job Stat Modifiers (per class)

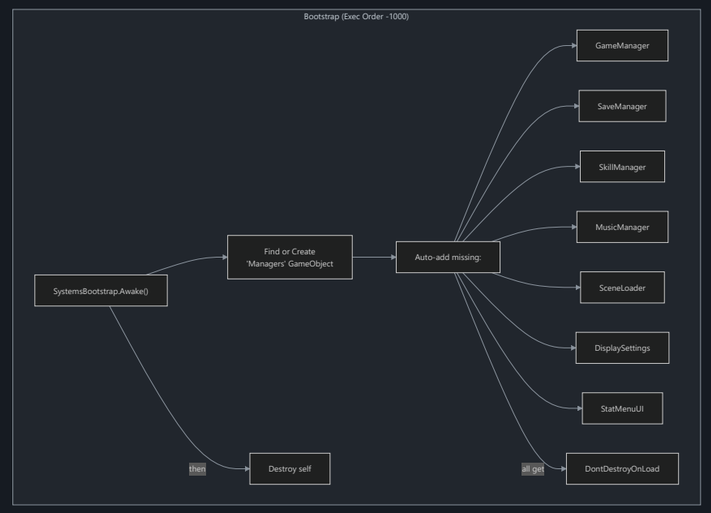
- **Warrior** — attackModifier, high strPerLevel, parry: ClassicParry
- **Mage** — magicModifier, high intPerLevel, parry: SpellMirror
- **Rogue** — attack speed + crit, high agiPerLevel, parry: ShadowStep

5. Core Systems & Bootstrap

Singletons, game state machine, save system, and initialization order.



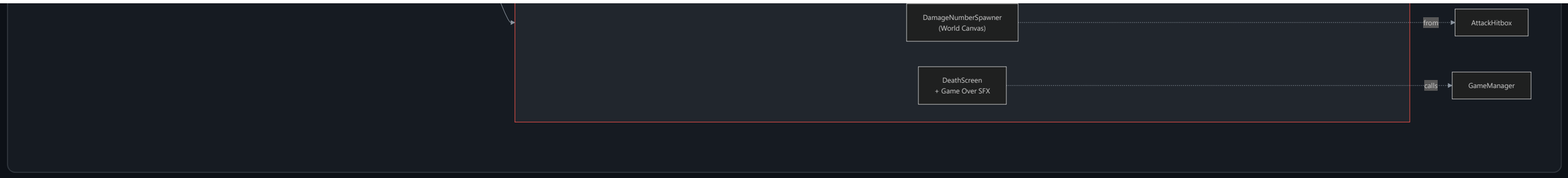




6. UI Layer

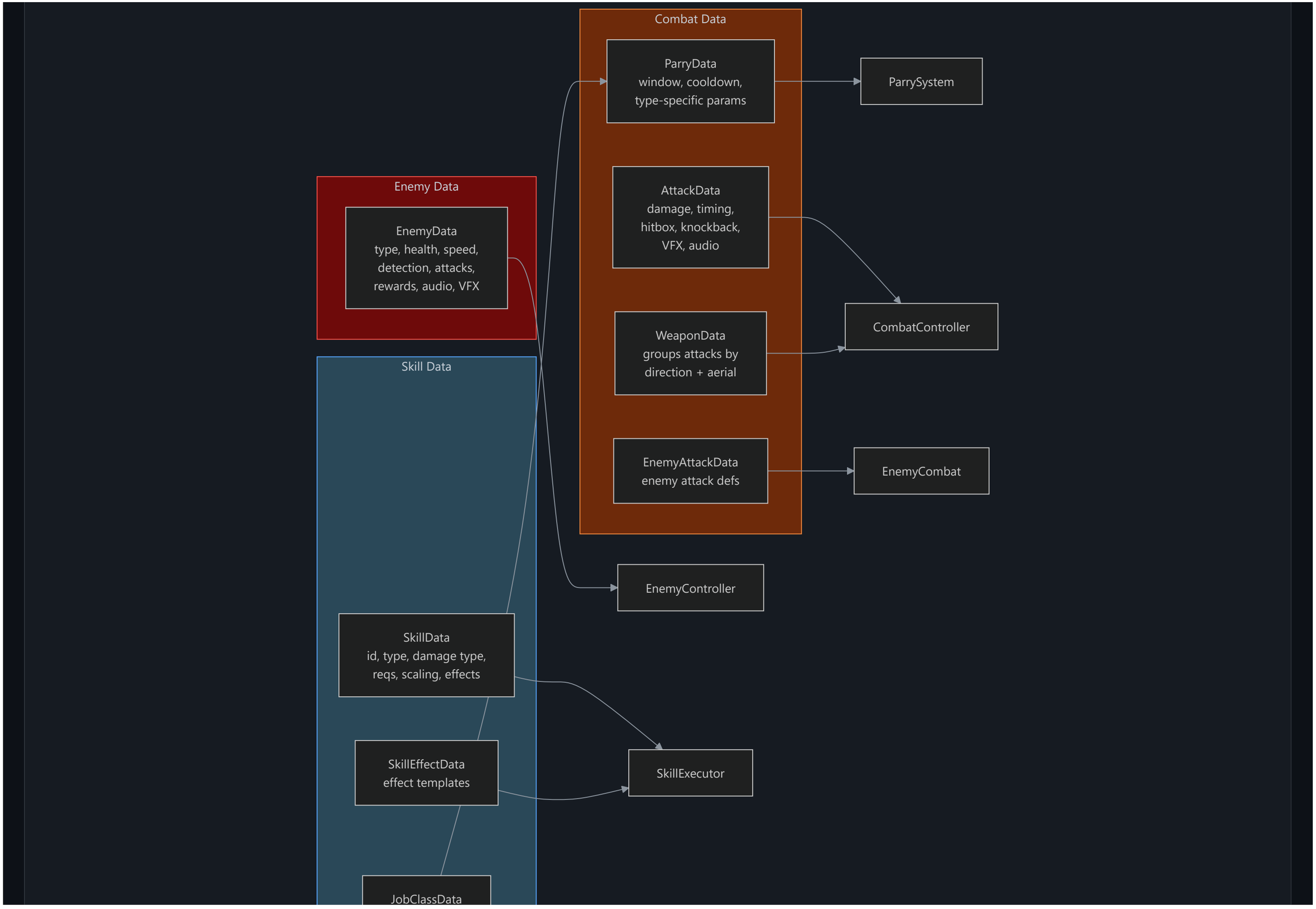
All UI screens, HUD elements, and their data sources.

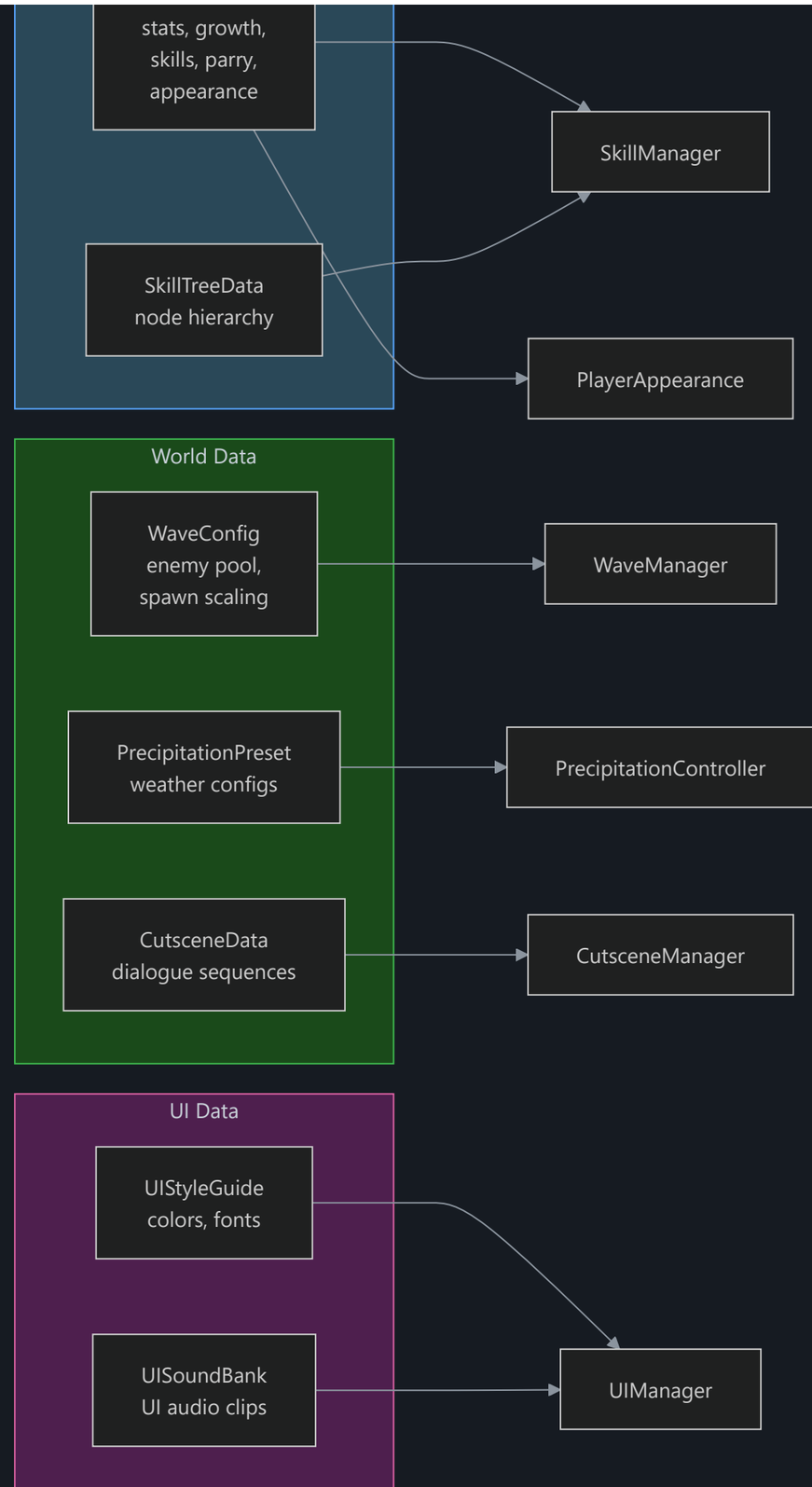
Menu Screens
MainMenuController
New/Continue/Options/Quit
CharacterCreation
Job select + appearance
PauseMenuController
OptionsMenu
Audio/Video/Controls
CreditsController
HighscoresController
SaveSlotUI
HUD Elements
PlayerStatBars
HP + MP bars
reads
HealthSystem
reads
ManaSystem
ExpBarDisplay
reads
LevelSystem
LevelDisplay
WeaponIndicator
CastBar
ComboCounter
BossHealthBar
reads
BossController
NotificationSystem
LowHealthVignette
UIManager (Singleton)
Canvas Refs:
mainMenu, hud,
pause, world
FocusManager
UIStyleGuide (SO)
UISoundBank (SO)
Skill & Stat UI
SkillTreeController
reads
SkillManager
SkillTreePanel
SkillNodeUI
SkillTooltip
ConnectionLines
SkillHotbar
JobAdvancementPopup
StatMenuController
STR/INT/AGI allocation
reads
StatSystem
Overlays


DamageNumberSpawner
(World Canvas)
from
AttackHitbox
DeathScreen
+ Game Over SFX
calls
GameManager

7. ScriptableObject Data Layer

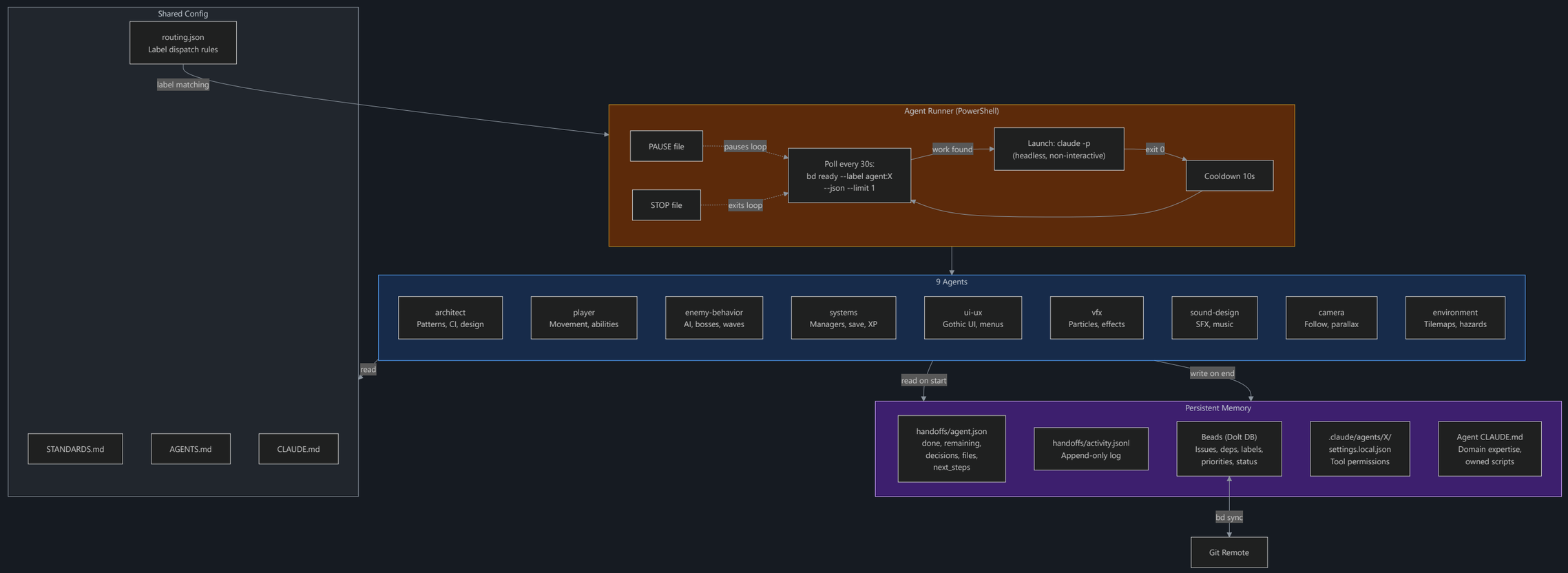
All data assets and what systems consume them.

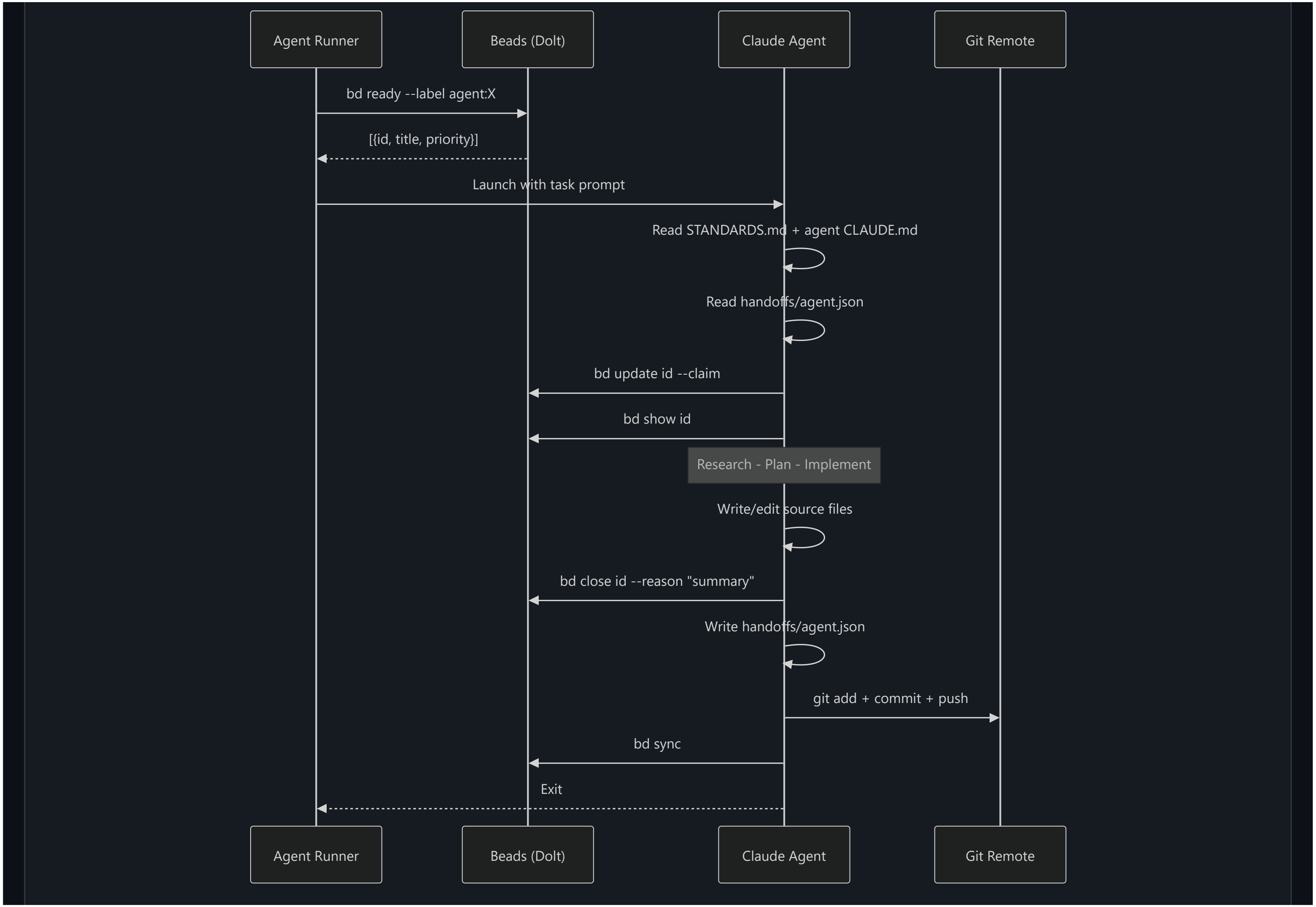

Combat Data
ParryData
window, cooldown,
type-specific params
ParrySystem
AttackData
damage, timing,
hitbox, knockback,
VFX, audio
Enemy Data
EnemyData
type, health, speed,
detection, attacks,
rewards, audio, VFX
WeaponData
groups attacks by
direction + aerial
CombatController
Skill Data
EnemyAttackData
enemy attack defs
EnemyCombat
EnemyController
SkillData
id, type, damage type,
reqs, scaling, effects
SkillEffectData
effect templates
SkillExecutor
JobClassData


stats, growth, skills, parry, appearance
SkillTreeData node hierarchy
SkillManager
PlayerAppearance
World Data
WaveConfig enemy pool, spawn scaling
WaveManager
PrecipitationPreset weather configs
PrecipitationController
CutsceneData dialogue sequences
CutsceneManager
UI Data
UIStyleGuide colors, fonts
UISoundBank UI audio clips
UIManager

8. Agent Infrastructure

How 9 autonomous Claude agents are dispatched, run, and persist context.




Agent Runner
Beads (Dolt)
Claude Agent
Git Remote
bd ready --label agent:X
[{id, title, priority}]
Launch with task prompt
Read STANDARDS.md + agent CLAUDE.md
Read handoffs/agent.json
bd update id --claim
bd show id
Research - Plan - Implement
Write/edit source files
bd close id --reason "summary"
Write handoffs/agent.json
git add + commit + push
bd sync
Exit
Agent Runner
Beads (Dolt)
Claude Agent
Git Remote

9. Current Work & Blocking Chain

What's in progress, what's blocked, and the critical path.

Completed (49)
1s2: SO Data Assets
P0 CLOSED
b8z: Input Consolidation
P0
...46 more closed beads
unblocked
In Progress (7)
9ag: Prefab Audit
P0 architect
i17: Multi-Scene
Zone System
P1 architect
17p: Unit Tests
P2 architect
buu: CI Pipeline
P2 architect
mn2: Inventory
P3 systems
bxi: Balance Curves
P3 systems
b8z: Input Assets
P0 player
blocks
blocks
blocks
blocks
blocks
blocks
Blocked (6)
1j4: Prefab CI Check
P2
6bd: Boss Encounters
P1
80c: Zone Enemy Groups
P1
r7p: Exploration Gates
P1
7hq: NPC/Dialogue
P3
4za: World Map UI
P3

Critical Path Analysis

i17 (Zone System) is the single biggest bottleneck — it blocks 5 of 6 blocked beads. It's assigned to the **architect** agent, who is also carrying 3 other in-progress tasks (prefab audit, unit tests, CI pipeline).

Recommendation: Prioritize i17 completion or break it into smaller, unblocking chunks. Consider moving architect's lower-priority work (17p, buu) to backlog to focus bandwidth.

10. Project File Structure

Scripts (155 files)

Directory	Count	Key Classes
Player/	2	PlayerControllerScript, PlayerAppearance
Abilities/	3	DashAbility, DoubleJumpAbility, PowerUpManager
Combat/	~12	CombatController, ParrySystem, AttackHitbox, Projectile
Enemies/	~20	EnemyController, BossController, Movement variants, Sensors
Skills/	~15	SkillManager, SkillExecutor, Effects (4 types), Trackers
Systems/	~15	GameManager, SaveManager, Health/Mana/Level/StatSystem
UI/	~35	UIManager, all menus, HUD, skill tree, stat menu
Camera/	5	AdvancedCameraController, Parallax, Bounds, BossRoom
Audio/	4	SFXManager (static), MusicManager (singleton)
VFX/	~18	Death/Spawn/Hurt/Dash VFX, Fog, Precipitation
Editor/	~10	Setup wizards, build script, debug tools

Prefabs

Directory	Contents
Player/	Player.prefab
Enemies/	8 enemies + GuardianBoss
Abilities/	DashPowerUp, DoubleJumpPowerUp
Effects/	Death/Spawn/Skill VFX, XP Orb, 6 Parallax layers
UI/	HUD, MainMenu, SkillTree canvases + UIManager
Camera/	MainCamera
Arena/	SurvivalArena

Scenes

MainMenu.unity	Title screen
SampleScene.unity	Gameplay (single scene currently)

Packages

Input System	1.17.0
Universal RP	17.3.0
Cinemachine	3.1.5
2D Suite	Sprites, Tilemap, Animation, SpriteShape, Aseprite